

Intro to DSP - MUSIC 320 //Homework 4 (100 Points)

Handed out on: November 3th, 2025 - 3:30 pm

Due on: November 17th, 2025 - 1:30 pm

Submission instructions

The homework must be submitted using Canvas's assignment submission system (in the Assignments tab). Select the associated link in that tab and proceed with your submission.

You must **turn in 2 items**:

- **A write-up (with all the instructed answers and figures)** in PDF format. Scanned and typed submission are both acceptable.
- **All your code and data files in a single compressed file** (zip, tar or tar.gz) containing the jupyter notebook with all the answers to the lab exercises and any other files requested in the homework instructions.

For your files, use the following **naming convention**: hwX.suid.type where X is the homework number and suid is your Stanford ID. For example, the write-up for homework 1 will be hw1_mab.pdf and the code files hw1_mab.zip/hw1_mab.tar.gz/hw1_mab.tar.

Theory Problems (60 pts.)

1. (20 pts) Convolution and Correlation:
The *impulse* or "unit pulse" signal is defined by

$$\delta[n] \triangleq \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

For example, $\delta[n] = [1, 0, 0, 0]$ for $N = 4$.

- (a) Verify that the impulse signal is the *identity element* under convolution using the impulse signal $\delta = [1, 0, 0, 0]$ and the input signal $x = [1, -1, 1, -1]$. That is, show that $x \circledast \delta = x$.
- (b) Show that $x \circledast \text{Shift}_1(\delta) = \text{Shift}_1(x)$, where $\text{SHIFT}_{1,n}(x) \triangleq x[n-1]$.
- (c) Find $(x \circledast [1, 1, 0, 0, \dots])_n$ using linearity of the convolution operator and the results above.
- (d) For the signals:

$$\begin{aligned} x &= [0, 0, 1, 0, 0, 0, 1, 0] \quad \text{and} \\ h &= [1, 1, 1, 1, 1, 0, 0, 0], \end{aligned}$$

$$\text{find } y[n] = (x \circledast h)_n$$

- (e) For $x = [1, 2, 3, 2]$ and $h = [3, -1, 2, 1]$, find the convolution $(x \circledast h)_n$ and correlation $(x \star h)_n$. Note that they are both cyclic. Hint: knowing x is even can help you.

2. (15 pts.) Convolution Theorem:

Consider the signal $x[n] = \cos(n\pi/2)$ and a rectangular window $w[n]$ which is equal to 1 for $0 \leq n < 4$ and zero elsewhere. We will consider the time period $0 \leq n < 8$.

- Compute the $N = 8$ DFT of $x[n]$, $w[n]$,
- Compute the $N = 8$ DFT of their product $xw[n] = x[n]w[n]$.
- Use these DFT results to show that $XW[k]$ is $1/N$ times the convolution of $X[k]$ and $W[k]$.
- Compute the $N = 8$ IDFT of the convolution of $X[k]$ and $W[k]$.

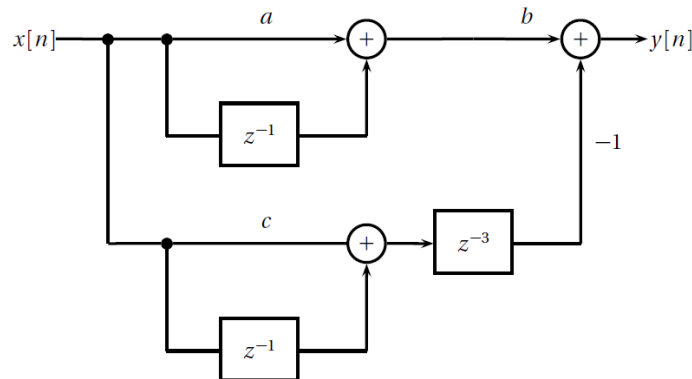
3. (15 pts.) LTI, FIR/IIR Filters:

Show that the following filters are either linear or non-linear, time invariant or time varying; also classify the filters as either FIR or IIR:

- $y[n] = x[n] + 2x[n-1] + x[n-2]$
- $y[n] = x[n] - y[n-1]$
- $y[n] = x[n] - 0.5nx[n-1]$
- $y[n] = 3\sqrt{x[n]}$

4. (10 pts.) Difference Equation:

Write the difference equation for the following filter:



Lab Assignments (40 Points) Please see the attached Jupyter notebook.